Instagram Analytics Report

An End-to-End Data Analytics Project

Python · Power Query · Power BI

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Abstract

This project presents a comprehensive end-to-end data analytics solution for understanding Instagram content performance across diverse categories, media types, and traffic sources. Using a dataset of 30,087 post records spanning 15 attributes and covering the period November 2024 to November 2025, the project applies Python-based data cleaning and preprocessing techniques, followed by advanced data transformation using Power Query. Key insights are surfaced through an interactive Power BI dashboard featuring KPI cards, trend charts, category breakdowns, and dynamic filtering capabilities. The analysis reveals meaningful patterns in content engagement, revenue generation across categories, media type performance, and traffic source distribution — all of which carry significant implications for content strategy and digital marketing decisions.

1. Problem Statement

In today's highly competitive social media landscape, understanding Instagram content performance is no longer a luxury — it is a strategic necessity. Content creators and brands collect vast amounts of engagement data, yet most of it remains unused due to the lack of structured analytical pipelines. This project addresses the following core business problems:

- Which content categories and media types generate the most revenue and highest engagement?
- How do different traffic sources (Explore, Hashtags, Reels Feed, Home Feed, Profile, External) distribute revenue and reach?
- What are the temporal trends in revenue and profit across the year 2024–2025?
- How does media type (Video, Reel, Carousel, Photo) influence revenue performance?
- What engagement patterns emerge across different content categories?
- How can data quality issues such as inconsistent labels, missing values, and outliers be systematically resolved?

The absence of structured analysis leads to suboptimal content planning, poor budget allocation, and missed growth opportunities. This project provides a replicable, data-driven framework to answer these questions systematically.

2. Solution & Features

2.1 Data Pipeline

The project implements a complete data analytics pipeline spanning four key stages:

Stage	Description	Tool
Data Loading	Loaded raw CSV into Pandas DataFrame for initial exploration	Python / Pandas
EDA & Profiling	Dtype checks, null analysis, duplicate detection, distribution summaries	Pandas
Data Cleaning	Standardized labels, replaced sentinel values, handled missing data, removed duplicates, treated outliers using IQR method	Python / Pandas
Data Transformation	Further normalization of categorical values, type conversions, final validation	Power Query (Power BI)
Visualization	Interactive dashboard with KPI cards, trend lines, donut/bar charts, and slicers	Power BI Desktop

2.2 Key Features

Systematic Null Imputation:

Numeric columns (likes, comments, shares, saves, reach, impressions, etc.) were imputed using column-wise median values after outlier removal — a principled approach that preserves distribution integrity and avoids cross-group bias.

IQR-Based Outlier Treatment:

An automated IQR function was implemented to detect and nullify extreme values across all 10 numeric columns. This ensures statistical validity without permanently discarding records.

Label Standardization:

Inconsistent categorical labels were systematically corrected — for example, 'photo' → 'Photo', 'VIDEO' → 'Video', 'Carousal' → 'Carousel', 'home feed' → 'Home Feed', 'tech' → 'Technology'. This ensures accurate grouping in analysis.

Sentinel Value Replacement:

Values such as 'unknown', 'N/A', '-', 'missing', '%error', 'not available', and 'None' were replaced with proper NaN values before imputation — preventing corrupted data from skewing results.

Interactive Dashboard:

The Power BI dashboard features KPI cards for Revenue (3bn), Cost (151M), Profit (3bn), Profit Margin (0.95), and Total Reach (30bn), with dynamic slicers for traffic_source and upload_date range filtering.

3. Technology Stack

Layer	Technology	Purpose
Programming	Python 3.x	Data loading, EDA, cleaning, and feature standardization
Data Manipulation	Pandas / NumPy	DataFrame operations, transformation, imputation, and aggregation
Data Transformation	Power Query (M Language)	Further categorical cleaning, type enforcement, and table shaping
Visualization	Power BI Desktop	Interactive dashboard with slicers, KPI cards, and charts
Notebook Environment	Google Colab (Jupyter)	Reproducible analytics and cleaning pipeline
Version Control	Git & GitHub	Source control and public project hosting

4. Dataset Analysis

4.1 Dataset Summary

Attribute	Value
Total Records (after cleaning)	30,087
Total Features	15
Date Range	19 November 2024 – 19 November 2025
Media Types	Video, Reel, Carousel, Photo
Traffic Sources	Hashtags, Home Feed, Reels Feed, External, Profile, Explore
Content Categories	Technology, Photography, Fashion, Lifestyle, Food, Fitness, Music, Beauty, Travel, Comedy
Avg Engagement Rate	9.76%
Total Reach	~30 Billion

4.2 Media Type Distribution

Media Type	Record Count	% of Total	Revenue Contribution

Video	8,005	26.6%	0.79bn (Highest)
Carousel	7,397	24.6%	0.73bn
Reel	7,387	24.5%	0.74bn
Photo	7,298	24.3%	0.73bn

4.3 Content Category Distribution

Category	Record Count	Revenue Contribution
Technology	3,592	0.36bn (Highest)
Photography	2,980	0.30bn
Fashion	2,975	0.30bn
Lifestyle	2,970	0.29bn
Food	2,958	0.29bn
Music / Comedy / Others	~2,900 each	~0.29bn each

5. Key Insights & Findings

1. Technology Category Dominance

Technology content generates the highest revenue at 0.36bn, accounting for a disproportionate share despite being only ~12% of total records. This indicates higher monetization potential per post in the tech category.

2. Video Leads in Revenue by Media Type

Video posts contribute 0.79bn in revenue, outperforming Reels (0.74bn), Carousels (0.73bn), and Photos (0.73bn). The margin is modest, suggesting all media types perform comparably, but video has a slight edge for revenue generation.

3. Balanced Traffic Source Distribution

Revenue is distributed almost equally across all six traffic sources, with Hashtags at 18.23%, Explore at 15.91%, External at 16.49%, Home Feed at 16.44%, Reels Feed at 16.56%, and Profile at 16.36%. This indicates a well-diversified traffic mix with no single source dependency.

4. Revenue vs. Profit Trend Spike

The Revenue vs. Profit trend chart reveals a significant spike around June 2025, suggesting a major campaign, viral content event, or algorithm boost during that period. Outside this event, revenue remains stable and consistent across the year.

5. High Profit Margin

A profit margin of 0.95 (95%) indicates extremely high-efficiency monetization relative to operational costs. Revenue of 3bn against a cost of only 151M points to a scalable, low-cost content operation.

6. Categorical Label Inconsistency (Data Quality Finding)

A programmatic audit revealed 15+ distinct label variations across media_type, traffic_source, and content_category columns (e.g., 'Carousal' vs 'Carousel', 'EXPLORE' vs 'Explore', 'home feed' vs 'Home Feed'). Resolving these was critical to ensure accurate groupings in the dashboard.

7. Outlier Presence in Engagement Metrics

The IQR-based outlier detection identified extreme values in all 10 numeric columns, including abnormally high reach and impressions values. These were set to NaN and replaced with column medians to maintain analytical integrity.

8. Uniform Engagement Across Categories

Average engagement rate of 9.76% is consistent across categories, with no single category significantly outperforming others. This suggests that content quality and timing matter more than category choice for engagement.

6. Power BI Dashboard

The Power BI dashboard presents all analytical findings in an interactive, business-ready format. It is designed for non-technical stakeholders and enables self-service exploration through dynamic slicers for traffic source and upload date range.

Figure 1: Instagram Analytics Power BI Dashboard

Visual Component	Metric / Dimension	Business Use
KPI Cards (5)	Revenue, Cost, Profit, Profit Margin, Total Reach	Snapshot of overall performance at a glance
Line Chart	Revenue vs. Profit Trend over time	Identifies seasonal patterns and campaign impact
Bar Chart	Revenue by Content Category	Guides content investment decisions by category
Donut Chart	Revenue by Traffic Source	Reveals channel effectiveness and traffic diversity
Horizontal Bar Chart	Revenue by Media Type	Compares monetization across post formats
Slicer Panel	Traffic Source filter	Dynamic segment-level filtering for custom analysis
Date Slicer	Upload Date range selector	Time-period analysis and trend isolation

7. Unique & Distinguished Points

- End-to-end analytics pipeline: From raw unclean CSV through Python EDA and cleaning, Power Query transformation, to a fully interactive Power BI dashboard — all reproducible via a single Colab notebook.
- Dual-layer cleaning approach: Python handles null imputation, outlier treatment, and label standardization; Power Query performs a second pass for final normalization — ensuring maximum data quality before visualization.
- IQR-based automated outlier detection: A reusable treat_outliers_iqr() function was applied across all 10 numeric columns, eliminating extreme values without permanently losing records.
- Sentinel value inventory: Systematically identified and replaced 8 distinct sentinel value patterns ('unknown', 'N/A', '-', 'missing', '%error', 'not available', 'nan', 'None') that would otherwise corrupt aggregations.
- Category-median imputation strategy: Numeric missing values were filled using per-column median (not mean), providing outlier-resistant imputation that preserves the distribution shape.
- Real-time interactive filtering in Power BI: Slicers for traffic source and date range allow any business user to explore the full dataset across multiple dimensions without writing code.
- Comprehensive label audit: 15+ label inconsistency variants were identified and corrected programmatically, ensuring that groupings in the dashboard are accurate and trustworthy.

8. Future Improvements

Content Performance Prediction (ML):

Train a regression model (Random Forest / XGBoost) to predict post engagement rate based on media type, category, hashtag count, caption length, and upload timing — enabling proactive content optimization.

Audience Segmentation:

Apply K-Means or DBSCAN clustering on engagement metrics to identify distinct audience behavior segments, enabling hyper-targeted content campaigns.

Sentiment Analysis on Captions:

Apply NLP sentiment analysis on post captions to correlate caption tone (positive, neutral, negative) with engagement outcomes and revenue generation.

Real-time Dashboard:

Integrate the Instagram Graph API or a live data pipeline (e.g., Azure Event Hub) to feed the Power BI dashboard with real-time metrics for live campaign monitoring.

Hashtag Effectiveness Analysis:

Analyze hashtag count versus reach and engagement to determine the optimal number of hashtags per post type and category.

Geospatial Analysis:

Incorporate audience geography data to map engagement and follower growth across regions using Power BI's filled map visual or Python's Folium library.

This project was independently developed by Tanmay Tiwari (Roll: 2328055) as part of the B.Tech CSE curriculum. All analysis, code, and documentation are original work.

GitHub: github.com/tanmay218/Instagram_Analysis